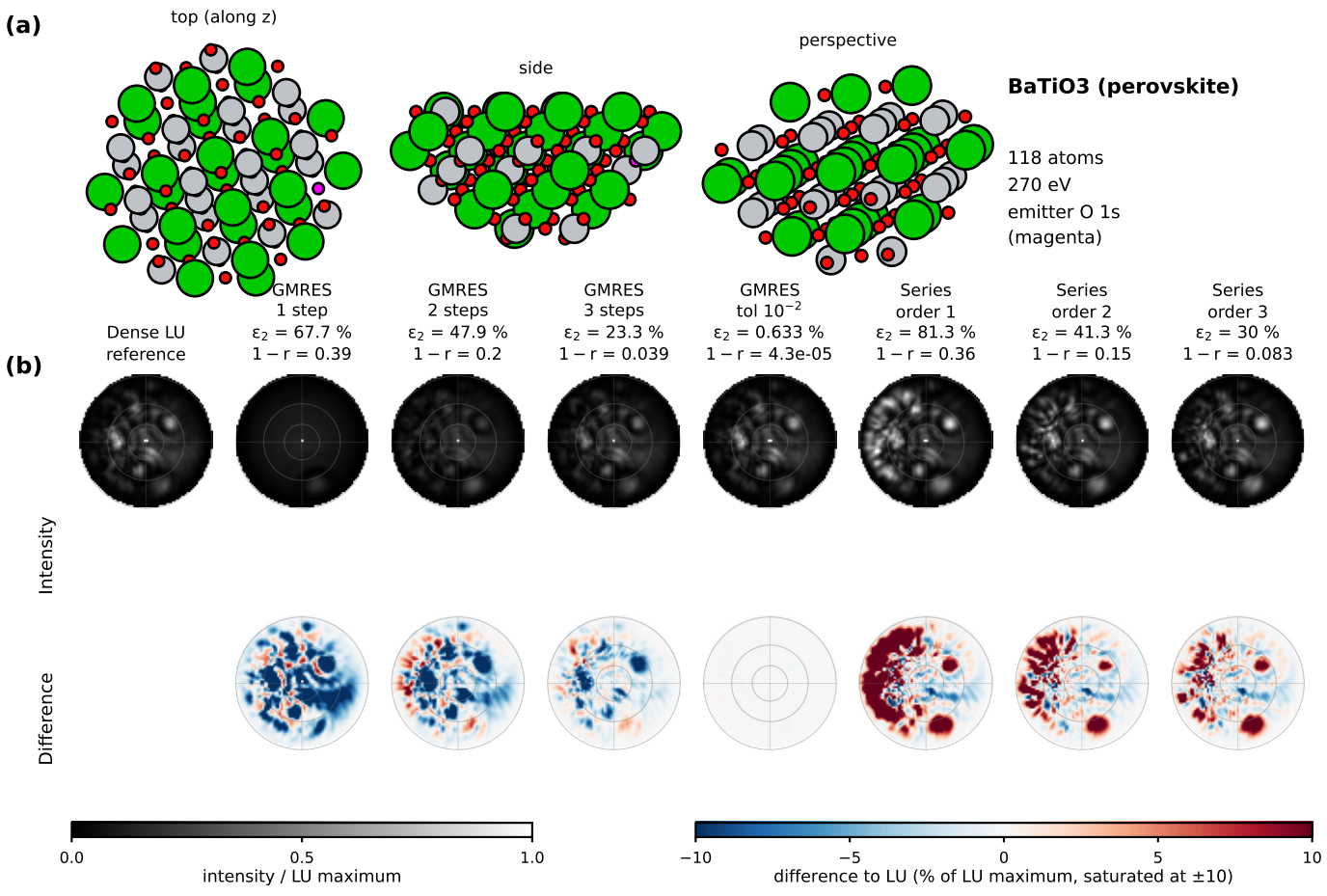

(a)
top (along z)
side
perspective
BaTiO3 (perovskite)
118 atoms
270 eV
emitter O 1s
(magenta)
(b)
Dense LU reference
GMRES 1 step
ε2 = 67.7 %
1 − r = 0.39
GMRES 2 steps
ε2 = 47.9 %
1 − r = 0.2
GMRES 3 steps
ε2 = 23.3 %
1 − r = 0.039
GMRES tol 10−2
ε2 = 0.633 %
1 − r = 4.3e-05
Series order 1
ε2 = 81.3 %
1 − r = 0.36
Series order 2
ε2 = 41.3 %
1 − r = 0.15
Series order 3
ε2 = 30 %
1 − r = 0.083
Intensity
Difference
0.0
0.5
1.0
intensity / LU maximum
−10
−5
0
5
10
difference to LU (% of LU maximum, saturated at ±10)